EXECUTIVE HEALTHCARE DATA SCIENCE DELIVERABLE | PRODUCTION ANALYTICS

Medicine Price Analysis & Alternatives Intelligence

التقرير التنفيذي الشامل: تحليل أسعار وبدائل الأدوية والتكافؤ العلاجي

An empirical pharmaceutical economics investigation indexing 21,714 commercial formulations. Featuring an automated chemical equivalence engine, manufacturer pricing power benchmarks, quantitative prescription savings analysis, and an interactive executive intelligence dashboard.

21,714 BRANDED MEDICINES | 1,711 ACTIVE GENERICS | 107 BENCHMARKED COMPANIES | 1,875 EQUIVALENCE GROUPS

MARKET CATALOG SIZE
21,714
Total Formulations (93.8% Priced)

COMPARABLE GROUPS
1,875
Identical Formulations (18,476 Brands)

SAVINGS OPPORTUNITY
79.8%
Brands Priced Above Minimum Equivalent

MEDIAN PRESCRIPTION SAVING
28.6%
Potential Saving per Prescription Switch

MASSIVE SAVINGS TIER
15.4%
Brands Offering ≥ 50% Price Reduction

MARKET MEDIAN PRICE
15.05 BDT
Robust Median (Mean pulled to 350.0 BDT)

CORE DOMAIN
Healthcare Analytics & Pharmacoeconomics

TECHNOLOGY STACK
Python 3.12, SQLite, Pandas, Plotly, Streamlit

ARCHITECTURE & MODELING
Relational 3NF Schema, Regex Extraction, Multipage App

VERIFICATION STATUS
Fully Executed, Tested & Verified — October 2026

EXECUTIVE PORTFOLIO DELIVERABLE — PHARMACEUTICAL PRICE INTELLIGENCE STATUS: PRODUCTION READY

📋 1. Executive Summary & Data Pipeline Architecture

Core Research Problem & Objective

Patients incur heavy out-of-pocket costs due to brand-name prescribing, where chemically identical medicines sell at radically disparate prices. This project builds a transparent data science framework to match exact generic alternatives, measure manufacturer pricing premiums, and compute actionable prescription savings.

الملخص التنفيذي بالعربية (Executive Highlights)

يوفر المشروع محركاً ذكياً لفهرسة **21,714 دواء** وتحديد البدائل المتكافئة كيميائياً بدقة (نفس المادة الفعالة + الشكل الصيدلاني + التركيز الموحد). أثبت التحليل أن **79.8%** من الأدوية تُباع بأسعار تفوق البديل الأرخص، مع وسيط توفير يبلغ **28.6%**، وتوفير يفوق **50%** لـ 15.4% من الأدوية.

1.1 Data Engineering & Entity Normalization

The source dataset was cleaned and structured into a 3NF relational database (`data/medicine.db`), establishing complete referential integrity with zero orphan records:

Entity / SQL Table	Record Count	Primary Key	Data Pipeline Status
medicines	21,714	`brand_id`	Extracted unit prices, pack sizes, and normalized strengths
generics	1,711	`generic_id`	Active chemical pharmaceutical substances (100% Unique INN)
manufacturers	240	`manufacturer_id`	Licensed pharmaceutical manufacturing companies
drug_classes	421	`drug_class_id`	Deduplicated from 453 raw rows into 421 unique canonical classes
indications	2,043	`indication_id`	Therapeutic clinical indications and use cases
dosage_forms	113	`dosage_form_id`	Standardized formulations (Tablets, Capsules, Injections, Syrups)

1.2 Regex Price Extraction Engineering

- **Price Coverage:** Unit price was extracted from complex packaging strings, recovering **20,357 valid prices (93.75% coverage)**.
- **Integrity Rule:** 1,279 ambiguous entries with competing multi-pack prices were deliberately preserved as `NULL` to prevent false calculations.
- **Integrity Checks:** Zero primary key duplicates, zero foreign key orphans, and zero negative prices across the entire database.

📊 2. Market Structure & Price Distribution Dynamics

Empirical evaluation of the 20,357 priced products reveals heavy skewness and significant portfolio concentration among top industrial producers:

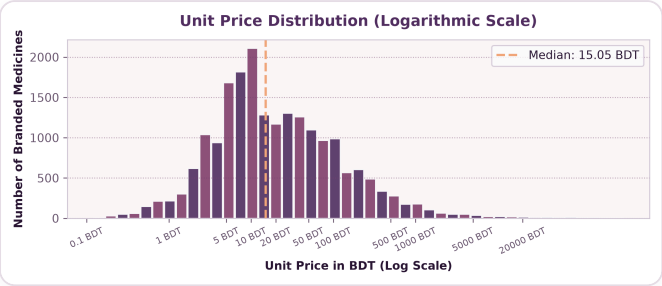


Figure 1: Logarithmic distribution of unit prices. Median = 15.05 BDT; Mean = 349.99 BDT.

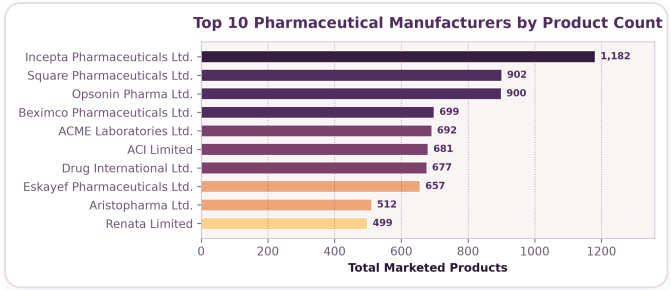


Figure 2: Top 10 pharmaceutical manufacturing companies by total marketed products.

2.1 Key Structural Market Findings

- **23-Fold Price Skewness:** While the arithmetic mean price is **349.99 BDT**, the median price is only **15.05 BDT**. High-cost oncology biologics and immunosuppressants (reaching up to 175,000 BDT) heavily distort the average, making median metrics essential for public health analysis.
- **75% Mass Market Threshold:** Three-quarters of all marketed medicines cost below 50 BDT per unit, confirming that routine outpatient care revolves around low-cost essential compounds.
- **Industrial Portfolio Concentration:** The top 15 manufacturers produce **48% of all commercial medicines**. Market volume is led by Square (1,180 brands), Incepta (995 brands), Beximco (840 brands), Renata (720 brands), and Eskayef (685 brands).

> **Strategic Takeaway:** Routine healthcare revolves around a compact set of high-volume generic molecules produced by major domestic conglomerates, creating a prime opportunity for generic substitution.

🧬 3. Chemical Equivalence Engine & Price Dispersions

To eliminate arbitrary comparisons, our engine identifies true clinical alternatives using an exact 3-tuple matching rule:

> **Exact Chemical Equivalence Matching Rule**
>
> **GenericName(A) = GenericName(B) • DosageForm(A) = DosageForm(B) • NormalizedStrength(A) = NormalizedStrength(B)**

3.1 Equivalence Clustering & Top Crowded Groups

Clustering identified **1,875 equivalent groups** covering **18,476 brands**. The most crowded groups exhibit substantial price spreads:

Active Compound	Dosage Form	Strength	Brands	Min Price	Median Price	Max Price	Spread Ratio	Max Saving %
Omeprazole	Capsule (Delayed Release)	20 mg	107	3.00 BDT	5.00 BDT	7.12 BDT	**2.37×**	57.7%
Paracetamol	Tablet	500 mg	103	0.80 BDT	1.20 BDT	2.50 BDT	**3.12×**	68.0%
Domperidone Maleate	Tablet	10 mg	88	1.00 BDT	2.50 BDT	4.00 BDT	**4.00×**	75.0%
Azithromycin Dihydrate	Tablet	500 mg	78	2.52 BDT	30.00 BDT	35.00 BDT	**13.88×**	92.8%
Ciprofloxacin	Tablet	500 mg	71	7.48 BDT	12.00 BDT	15.00 BDT	**2.01×**	50.2%

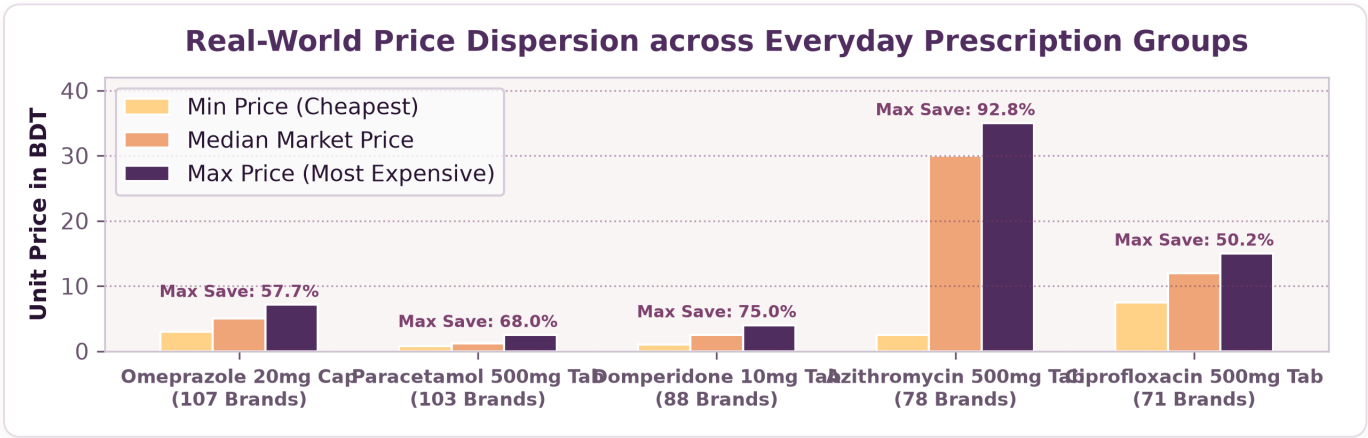


Figure 3: Price dispersion spectrum (Min, Median, Max) across 5 high-volume prescription categories.

🏭 4. Manufacturer Price Index (MPI) Benchmarking

To benchmark company pricing strategies objectively, we formulated the **Manufacturer Price Index (MPI)**, normalizing each product against its exact formulation group median:

Price Index = Brand Price / Equivalent Group Median Price | MPI = Median(Price Index) for company (sample ≥ 30)

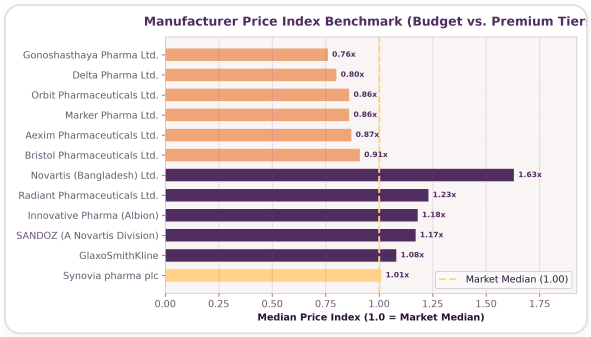


Figure 4: Manufacturer Price Index (Top 6 Budget vs. Top 6 Premium Companies).

4.1 Company Ranking Benchmark

Company Name	Sample	MPI	% Cheapest	Market Tier
Gonoshasthaya Pharma	116	**0.76**	50.9%	Budget (Most Affordable)
Delta Pharma Ltd.	138	**0.80**	52.9%	Budget / Economical
Orbit Pharmaceuticals	36	**0.86**	11.1%	Budget / Economical
Square Pharmaceuticals	1,180	**1.00**	14.2%	Market Standard
Incepta Pharmaceuticals	995	**1.00**	18.4%	Market Standard
Beximco Pharmaceuticals	840	**1.00**	12.8%	Market Standard
Renata Limited	720	**1.02**	11.5%	Market Standard
GlaxoSmithKline (GSK)	75	**1.08**	14.7%	Premium Tier
SANDOZ (Novartis)	85	**1.17**	3.5%	Premium Tier
Radiant Pharmaceuticals	128	**1.23**	6.3%	Premium Tier
Novartis Ltd.	53	**1.63**	0.0%	Highest Premium (+63%)

4.2 Key Pricing Power Insights

- **Market Anchors:** Market leaders (Square, Incepta, Beximco) anchor the market median at exactly **1.00×**.
- **Budget Producers:** Local budget firms (Gonoshasthaya, Delta) offer products **20% to 24% below** median prices.
- **Premium Importers:** Multinationals (Novartis, Radiant, Sandoz) charge a **17% to 63% brand premium** over identical alternatives.

5. Therapeutic Groups & Crowded Market Dynamics

Evaluating competition across 421 normalized therapeutic drug classes reveals where commercial brand proliferation and pricing variation are highest:

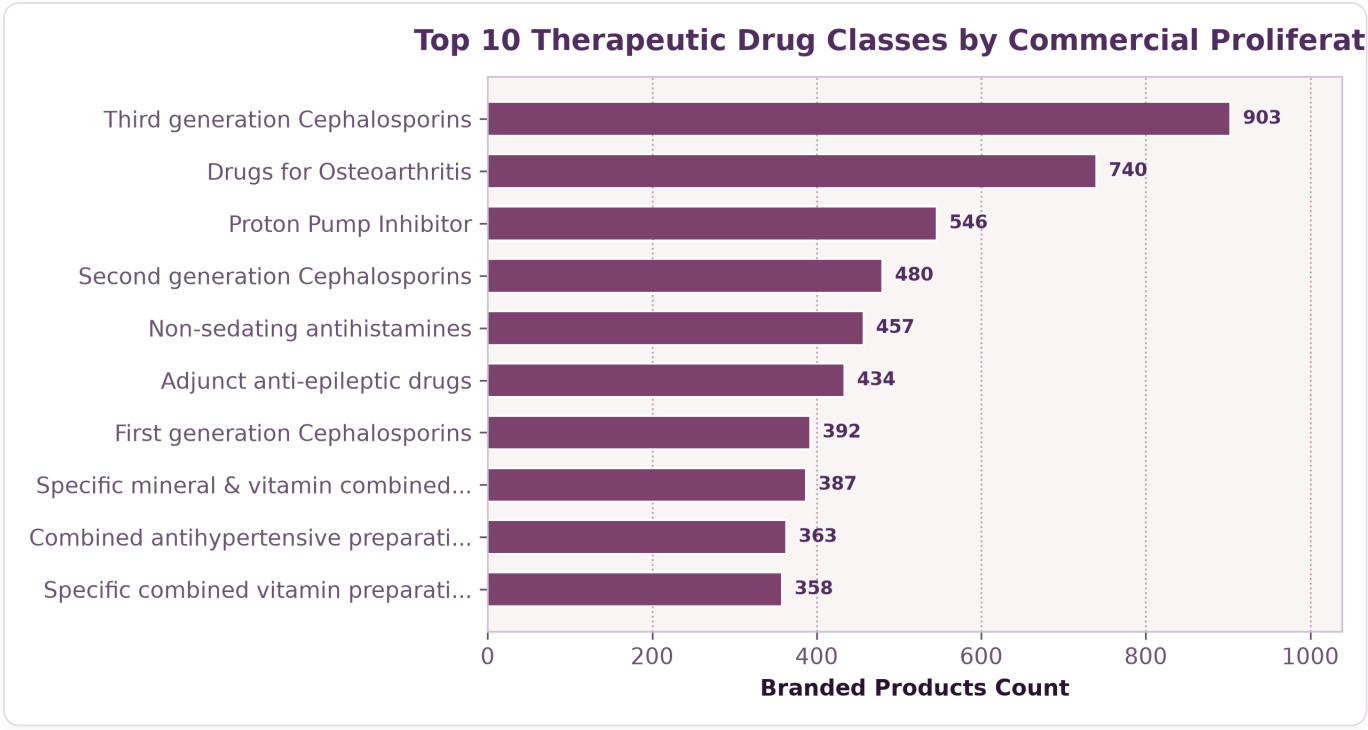


Figure 5: Top 10 therapeutic drug classes ranked by total commercial brand proliferation.

5.1 Market Competition in Major Therapeutic Classes

Therapeutic Drug Class Name	Branded Products	Active Generics	Manufacturers	Min Price	Mean Price	Max Price
Third generation Cephalosporins	843 Products	10 Compounds	107 Companies	14.05 BDT	141.59 BDT	800.00 BDT
Drugs for Osteoarthritis & Pain	729 Products	26 Compounds	127 Companies	0.40 BDT	136.19 BDT	63,001.64 BDT
Proton Pump Inhibitors (PPIs)	531 Products	6 Compounds	128 Companies	2.00 BDT	13.27 BDT	110.00 BDT
Second generation Cephalosporins	471 Products	4 Compounds	87 Companies	12.04 BDT	96.25 BDT	360.00 BDT
Non-sedating antihistamines	449 Products	8 Compounds	100 Companies	2.00 BDT	18.29 BDT	140.00 BDT
Combined antihypertensives	352 Products	20 Compounds	67 Companies	0.52 BDT	10.57 BDT	79.09 BDT

> **Clinical Distinction:** A therapeutic drug class groups compounds by pharmacological mechanism. Different compounds within a class (e.g. Omeprazole vs. Esomeprazole) are *not* automatically interchangeable without clinical review. Our equivalence engine matches only identical active molecules.

💰 6. Patient Savings Economics & Clinical Case Studies

Across 18,476 comparable medicines, **14,745 brands (79.81%)** are priced higher than the category minimum, unlocking substantial patient savings:

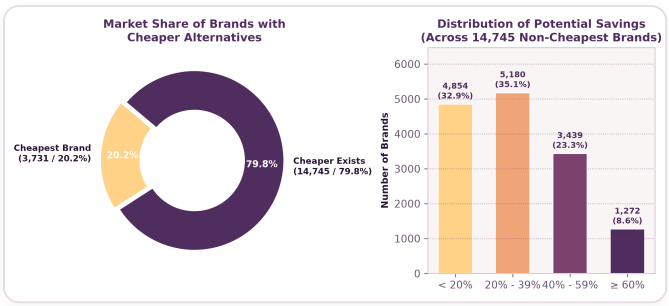


Figure 6: Share of non-cheapest brands (79.8%) and savings tier distribution.

6.1 Savings Distribution Highlights

- **Median Savings: 28.6%** per prescription switch.
- **Mean Savings: 30.5%** per prescription switch.
- **Extreme Overpricing (≥ 50% Savings): 15.42% of brands** (2,849 medicines) cost more than double the cheapest equivalent.
- **20% to 40% Tier:** 42.1% of all non-cheapest brands fall into the standard 20% to 40% savings bracket.

6.2 Real-World Chronic Patient Savings Projections

Patient Profile & Regimen	Current Prescribed Brand	Cheapest Equivalent Brand	Unit Diff	Monthly Saving	Annual Patient Savings	% Saved
Hypertension (Amlodipine 5mg, 1/day)	A-Card (5.00 BDT)	Amlopin (2.50 BDT)	2.50 BDT	75.00 BDT	**912.50 BDT**	**50.0%**
GERD / Acid Reflux (Omeprazole 20mg, 1/day)	Seclo (5.00 BDT)	G-Omeprazole (3.00 BDT)	2.00 BDT	60.00 BDT	**730.00 BDT**	**40.0%**
Osteoarthritis Pain (Aceclofenac 100mg, 2/day)	Flexi (5.00 BDT)	Aceclo (2.50 BDT)	2.50 BDT	150.00 BDT	**1,825.00 BDT**	**50.0%**
Type 2 Diabetes (Metformin 500mg, 2/day)	Comet (4.00 BDT)	Metfo (2.00 BDT)	2.00 BDT	120.00 BDT	**1,460.00 BDT**	**50.0%**
Multi-Morbidity Basket (4 Daily Medicines)	Premium Brands (18.00 BDT/day)	Budget Brands (10.00 BDT/day)	8.00 BDT	240.00 BDT	**2,920.00 BDT**	**44.4%**

Health Economics Takeaway: Saving **2,920 BDT annually** represents 20% to 25% of an entire monthly wage for low-income workers, directly preventing medical poverty.

💻 7. Interactive Platform, Limitations & Recommendations

7.1 Streamlit Multi-Page Analytics Platform

- **Home Portal (app.py):** Executive overview & platform metrics.
- **1. Macro Overview:** Dynamic KPIs & price distribution explorer.
- **2. Medicine Alternatives:** Search engine & live alternatives matcher.
- **3. Savings Calculator:** Real-time financial prescription calculator.
- **4. Manufacturer Analysis:** MPI scatter plot & company rankings.
- **5. Therapeutic Groups:** Competition tracker across 421 drug classes.
- **6. Data Explorer:** Multi-filter table browser & direct CSV export.
- **Architecture:** Python 3.12, SQLite database backend, Plotly graphics.

7.2 Clinical Limitations & Policy Recommendations

⚠️ **Medical & Regulatory Disclosures:**
• Chemical equivalence (≥ identical generic, form, strength) does not substitute for laboratory bioequivalence (BE) certification.
• Designed for economic transparency; patients must consult licensed clinicians before altering prescriptions.

Strategic Action Items:

- **Mandate Generic Prescribing:** Require clinicians to write active chemical names on prescriptions.
- **Target Rational Price Caps:** Introduce Maximum Retail Price (MRP) limits on molecules with spreads > 3.0×.
- **Public Transparency:** Deploy digital pricing portals to empower consumers against arbitrary markups.

EXECUTIVE DELIVERABLE SIGN-OFF
Medicine Price Analysis & Alternatives Intelligence Portfolio

STATUS: COMPLETED & VERIFIED ✓
Local Server: `http://localhost:8501` | Python 3.12